Apriori Algorithms

Market Basket algorithms

Association Rule based

- ✓ Min support = 50%
- ✓ Thresold confidence = 70%

Ex

TID	Items
101	1, 3, 4
102	2, 3, 5
103	1, 2, 3, 5
104	2, 5

Total = 4

I) Single point calculation

Itemset	support	
1	2/4 = 50%	2
2	3/4 = 75%	3
3	3/4 = 75%	3
X 4	1/4 = 25%	
5	3/4 = 75%	2

Since item no 4 has support < min support

25% < 50%

From here we continue with 1, 2, 3, 5 items only

(II) paired combination

Itemset	Support	
(1,2)	1/4 = 25% ✗	
(1,3)	2/4 = 50%	2
(1,5)	1/4 = 25% ✗	
(2,3)	2/4 = 50%	2
(2,5)	3/5 = 75%	3
(3,5)	2/4 = 50%	2

25% < 50%

(III) multiple combination

item set	support	
1, 3, 5	1/4 = 25	✗
2, 3, 5	2/4 = 50	
1, 2, 3	1/4 = 25	✗

✓ Support

(2, 3, 5) = 2

(IV) Generate Rules

Lift

Rules	Support
$(2 \wedge 3) \rightarrow 5$	2
$(3 \wedge 5) \rightarrow 2$	2
$(2 \wedge 5) \rightarrow 3$	2
$2 \rightarrow (3 \wedge 5)$	2
$5 \rightarrow (2 \wedge 3)$	2
$3 \rightarrow (2 \wedge 5)$	2

Calculate confidence –

$$\frac{S(A \cup B)}{S(A)}$$

Eg = $\underbrace{(2 \wedge 3)}_{A} \rightarrow \underbrace{5}_{B} = \frac{S[(2 \wedge 3) \cup 5]}{S(2 \wedge 3)}$

$$= \frac{2}{2}$$

$$= 100\%$$

Same for each point, then we got confidence

Confidence

① $2/2 = 100\%$ ✓

② $2/2 = 100\%$ ✓

③ $2/3 = 66\%$

④ $2/3 = 66\%$

⑤ $2/3 = 66\%$

⑥ $2/3 = 66\%$

we have confidence 70%

$\boxed{100 > 70}$

① $(2 \wedge 3) \rightarrow 5$

② $(3 \wedge 5) \rightarrow 2$

Both are having association b/w item